

Alip Arslan

571-526-8996 | Dallas, TX | aliparslan@outlook.com | github.com/aliparslan | linkedin.com/in/aliparslan | alip.dev

WORK EXPERIENCE

Capital One

Dallas, TX

Senior Associate Software Engineer, AI Research

July 2026 – Present

- Built a real-time message scoring service that automates approval of high-confidence AI chatbot responses across 50+ car dealerships, pairing an LLM judge with a similarity-search scorer to cut response latency by 45%
- Owned the routing logic, Snowflake analytics layer, and rollout for scoring service, validating scorer accuracy against human reviewers to move the system from shadow scoring to 70% of production traffic

Associate Software Engineer, AI Engineering

August 2025 – July 2026

- Built a Claude agent workflow that scans 15 components daily and opens automated fix PRs for 80+ vulnerabilities monthly
- Led continuous delivery initiative across 4 products for a 12-engineer team, reducing release cycle from 3-6 weeks to weekly via testing coverage gating across all environments, feature flags, and auto-merge deployment
- Co-invented a CLI harness that evaluates new model releases against a 60-conversation golden-dialog suite in parallel, cutting a full day of manual testing from 6 hours to 15 minutes; adopted by team PMs and partner R&D org
- Directed 14-person committee as Learning & Development Lead for 250+ associates, across 42 professional and networking events

Sheffield Lab of Computational Biology

Charlottesville, VA

Software Engineer (Research)

May 2022 – September 2024

- Standardized metadata for 250k genomic datasets from NIH by building a FastAPI/React validation tool (co-author, [GigaScience](#))
- Created benchmarking framework in Python to evaluate 22 transformer models for a vector-based search engine, improving query relevance by 37% and inference speed by 18%
- Accelerated GC-content calculation for DNA sequences by 12x via a tiling and caching system using pre-computed windows in Rust

Northrop Grumman

Charlottesville, VA

Software Engineer Intern

June 2023 – August 2023

- Engineered containerization strategy for a 2M+ line legacy navigation application for the Navy, enabling air-gapped Kubernetes deployment, zero-downtime updates, and multi-platform usability
- Led intern research optimizing C# codebase to resolve dependency bottlenecks, resulting in 6x faster application startup time

SKILLS

- **Languages:** Python, Java, TypeScript, JavaScript, SQL, HTML/CSS
- **Frameworks/Tools:** AWS, Spring Boot, Kafka, React, SolidJS, Kubernetes, Jenkins, PostgreSQL, Docker, Snowflake, Git
- **AI/ML:** PyTorch, LangGraph, LLM fine-tuning, RAG, embeddings & vector search, LLM-as-judge evaluation, shadow deployment
- **Certifications:** AWS Certified Solutions Architect (Associate)
- **Patents:** Co-inventor for two patents - LLM test automation for model evaluation, conversational quality scoring

PROJECTS

Pinkslip Jobs – Full Stack | Solo project | pinkslip.alip.dev

April 2026

- Built a job alerting pipeline on Cloudflare Workers that polls external sources, dedupes results, and pushes notifications on matches
- Shipped as an installable PWA or iOS app via a Capacitor wrapper, running entirely on edge infrastructure

Breadwinner – Full Stack | Solo project | breadwinner.alip.dev

November 2025

- Built a personal finance platform with bank-statement ingestion, automated transaction categorization, interactive spending visualizations, and monthly budget tracking
- Integrated Gemini Flash for spending insights; deployed serverless on Cloudflare Pages and Supabase

OneLearn – Full Stack | Team of 5

September 2025

- Built a peer-to-peer learning marketplace where engineers request, upvote, and host sessions on professional growth topics; placed 4th of 48 teams in Capital One's global hackathon
- Built the backend with FastAPI and PostgreSQL, with a React frontend for real-time event hosting and content voting

EDUCATION

University of Virginia

Charlottesville, VA

Master of Computer Science (MCS), GPA: 3.93

May 2025

BA in Computer Science, BA in Cognitive Science

May 2024